INCH-POUND

FF-S-86J
1 April 2025
SUPERSEDING
FF-S-86H
7 June 2021

FEDERAL SPECIFICATION
SCREW, CAP, SOCKET-HEAD

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1 SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers socket-head cap screws with multiple head configurations utilizing alloy steels, corrosion-resistant steels, heat and corrosion-resistant steel A286, non-ferrous materials including nickel-copper alloy, nickel-copper-aluminum alloy, brass, bronze, nickel alloys, or titanium alloys, with cadmium plating, zinc plating, zinc-nickel plating, phosphate coating or black chemical finish. See the appropriate sections for limitations on use of each material and coating.

1.2 Classification.

1.2.1 Types and sizes. Screws furnished under this specification should be of the following types (see 6.2 and figure 1). Screws should be classified according to size by the nominal body diameter as specified in ASME B18.3. Sizes covered range from .060 (#0) to 1.500 inches.

NOTE: Some of the parts covered by this document are for replacement purposes only, and may not be included in new designs.

Type I /1	-	Hexagon Socket (Cylindrical) Head Cap Screw 1936 Series
Type IV	-	Hexagon Socket Flat Countersunk Head Cap Screw 82˚ Included Angle
Type VI	-	Hexagon Socket (Cylindrical) Head Cap Screw
Type VIII	-	Hexagon Socket Button Head Cap Screw

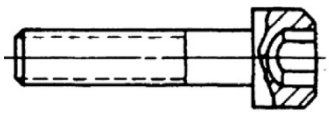
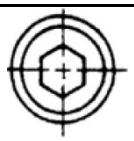
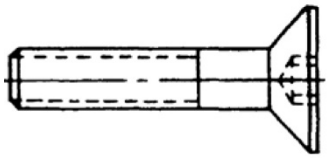
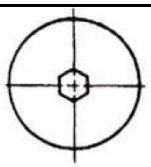
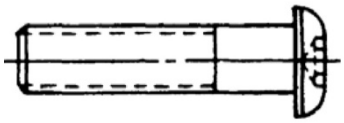
/1 DO NOT USE FOR NEW DESIGN (see 6.3.1).

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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Type	Configuration	Head
Type I /1 or VI /2	 CYLINDRICAL	
Type IV /2	 FLAT	
Type VIII /2	 BUTTON	

/1 Obsolete for new design (see 6.3.1). For procurement, it is recommended the design be evaluated for use with Type VI as replacement.
/2 Internal spline-drive fasteners should be replaced with hex configurations for Types IV, VI and VIII of the same head configuration.

Figure 1. Socket head screw types.

1.2.2 Part identification number system. A part identification number system should be used to identify each fastener specified in this document. Part identification may use the numbering system called out on a non-government or military standard (e.g., NAS1351-02-8 or NAS1351C04H12, or MS24672-1), or the supplier may optionally use ASME B18.24, but they must specify which part number identification system is being used, as noted in 6.2.

1.3 Terminology. Definitions of terms used in this specification are as specified in ASME B18.12, unless otherwise defined herein.

2 APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FEDERAL SPECIFICATIONS

QQ-N-281	-	Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip Wire, Forgings, and Structural and Special Shaped Sections
QQ-N-286	-	Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

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FEDERAL STANDARDS

FED-STD-H28/2	-	Screw-Thread Standards for Federal Services Section 2 Unified Inch Screw Threads- UN and UNR Thread Forms

MILITARY SPECIFICATIONS

MIL-DTL-13924	-	Coating, Inorganic, Black, for Ferrous Metals
MIL-DTL-16232	-	Phosphate Coating, Heavy, Manganese or Zinc Base
MIL-DTL-495	-	Finish, Chemical, Black, for Copper Alloys
MIL-PRF-32647	-	Zinc-Nickel Electroplating for Fasteners

MILITARY HANDBOOKS

MIL-HDBK-57	-	Listing of Fastener Manufacturers Identification Symbols

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.1.1 Other Government publications. The following other documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FEDERAL ACQUISITION REGULATION (FAR)

Paragraph 23.403	-	Emergency Planning and Toxic Release Reporting

(Copies of this document are available online at https://www.acquisition.gov/far/23.403.)

NAVAL SEA SYSTEMS COMMAND

T9074-AS-GIB-010/271	-	Requirements for Nondestructive Testing Methods

(Copies of this document are available from the Naval Sea Systems Command, Washington Navy Yard, DC 20376-5160.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS4003	-	Fastener, A286 Corrosion Resistant Alloy, Externally Threaded, 160 KSI Ftu, 95 KSI Fsu, 1000 °F
NAS4004	-	Fastener, 6AL-4V Titanium Alloy, Externally Threaded, 160 KSI Ftu, 95 KSI Fsu, 450 °F
NASM1312-13	-	Fastener Test Methods Method 13 Double Shear Test

(Copies of these documents are available online at www.aia-aerospace.org.)

AMERICAN PETROLEUM INSTITUTE (API)

API STD 6ACRA	-	Age-hardened Nickel-based Alloys for Oil and Gas Drilling and Production Equipment

(Copies of this document are available online at www.api.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes
ASQ H1607	-	Zero Acceptance Number Sampling Plans

(Copies of these documents are available online at www.asq.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1	-	Unified Inch Screw Threads (UN, UNR and UNJ Thread Forms)
ASME B1.3	-	Screw Thread Gaging Systems for Acceptability: Inch and Metric Screw Threads (UN, UNR, UNJ, M, and MJ)
ASME B18.3	-	Socket Cap, Shoulder, Set Screws, and Hex Keys (Inch Series)
ASME B18.12	-	Glossary of Terms for Mechanical Fasteners
ASME B18.18	-	Quality Assurance for Fasteners
ASME B18.24	-	Part Identifying Number (PIN) Code System Standard for B18 Fastener Products
ASME B46.1	-	Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of these documents are available online at www.asme.org.)

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ASTM INTERNATIONAL

ASTM A380/A380M	-	Standard Practice for Cleaning, Descaling, and Passivation of Stainless Steel Parts, Equipment and Systems
ASTM A493	-	Standard Specification for Stainless Steel Wire and Wire Rods for Cold Heading and Cold Forging
ASTM A574	-	Standard Specification for Alloy Steel Socket-Head Cap Screws
ASTM B21/B21M	-	Standard Specification for Naval Brass Rod, Bar, and Shapes
ASTM B98/B98M	-	Standard Specification for Copper-Silicon Alloy Rod, Bar and Shapes
ASTM B99/B99M	-	Standard Specification for Copper-Silicon Alloy Wire for General Applications
ASTM B138/B138M	-	Standard Specification for Manganese Bronze Rod, Bar, and Shapes
ASTM B139/B139M	-	Standard Specification for Phosphor Bronze Rod, Bar, and Shapes
ASTM B150/B150M	-	Standard Specification for Aluminum Bronze Rod, Bar, and Shapes
ASTM B348/B348M	-	Standard Specification for Titanium and Titanium Alloy Bars and Billets
ASTM B446	-	Standard Specification for Nickel-Chromium-Molybdenum-Niobium Alloy, Nickel-Chromium-Molybdenum-Silicon Alloy, and Nickel-Chromium-Molybdenum-Tungsten Alloy Rod and Bar
ASTM D3951	-	Standard Practice for Commercial Packaging
ASTM E8/E8M	-	Standard Test Methods for Tension Testing of Metallic Materials
ASTM E29	-	Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications
ASTM E1268	-	Standard Practice for Assessing the Degree of Banding or Orientation of Microstructures
ASTM E1282	-	Standard Guide for Specifying the Chemical Compositions and Selecting Sampling Practices and Quantitative Analysis Methods for Metals, Ores, and Related Materials
ASTM E1417/E1417M	-	Standard Practice for Liquid Penetrant Testing
ASTM E1444/E1444M	-	Standard Practice for Magnetic Particle Testing for Aerospace
ASTM F606/F606M	-	Standard Test Methods for Determining the Mechanical Properties of Externally and Internally Threaded Fasteners, Washers, Direct Tension Indicators, and Rivets

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ASTM F788/F788M	-	Standard Specification for Surface Discontinuities of Bolts, Screws, Studs, and Rivets, Inch and Metric Series
ASTM F835	-	Standard Specification for Alloy Steel Socket Button, Low and Flat Countersunk Head Cap Screws
ASTM F1941/F1941M	-	Standard Specification for Electrodeposited Coatings on Mechanical Fasteners, Inch and Metric

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL AEROSPACE ENVIRONMENTAL GROUP (IAEG)

AEROSPACE AND DEFENSE DECLARABLE SUBSTANCES LIST (AD-DSL)

(Copies of this document are available online at www.iaeg.com/chemicalrpt/addsl.)

SAE INTERNATIONAL

SAE AS1701	-	Lubricant, Solid Film
SAE AMS2461	-	Plating, Zinc-Nickel Alloy (12 to 16% Ni)
SAE AMS2488	-	Anodic Treatment - Titanium and Titanium Alloys Solution pH 13 or Higher
SAE AMS2631	-	Ultrasonic Inspection Titanium and Titanium Alloy Bar, Billet, and Plate
SAE AMS2700	-	Passivation of Corrosion Resistant Steels
SAE AMS2750	-	Pyrometry
SAE AMS2759/1	-	Heat Treatment of Carbon and Low-Alloy Steel Parts Minimum Tensile Strength Below 220 ksi (1517 MPa)
SAE AMS2759/3	-	Heat Treatment Precipitation-Hardening Corrosion-Resistant, Maraging, and Secondary Hardening Steel Parts
SAE AMS4928	-	Titanium Alloy Bars, Wire, Forgings, Rings, and Drawn Shapes 6Al - 4V Annealed - UNS R56400
SAE AMS4967	-	Titanium Alloy, Bars, Wire, Forgings, and Rings 6.0Al - 4.0V Annealed, Heat Treatable - UNS R56400
SAE AMS5662	-	Nickel Alloy, Corrosion- and Heat-Resistant, Bars, Forgings, Rings, and Stock for Forgings and Rings 52.5Ni - 19Cr - 3.0Mo - 5.1Cb (Nb) - 0.90Ti - 0.50Al - 18Fe Consumable Electrode or Vacuum Induction Melted 1775 °F (968 °C) Solution Heat Treated, Precipitation-Hardenable - UNS N07718
SAE AMS5726	-	Steel, Corrosion- and Heat-Resistant, Bars and Wire 15Cr - 25.5Ni - 1.2Mo - 2.1Ti - 0.006B - 0.30V 1800 °F (982°C) Solution Heat Treated and Work-Strengthened Consumable Electrode Remelted 200 ksi (1379 MPa) Tensile Strength Capability - UNS S66286

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SAE AMS5731	-	Steel, Corrosion- and Heat-Resistant, Bars, Wire, Forgings, Mechanical Tubing, Rings, and Stock for Forgings, Rings, or Heading 15Cr - 25.5Ni - 1.2Mo - 2.1Ti - 0.006B - 0.30V (A286) Consumable Electrode Remelted, 1800 °F (982 °C) Solution Heat Treated Precipitation Hardenable - UNS S66286
SAE AMS5732	-	Steel, Corrosion- and Heat-Resistant, Bars, Wire, Forgings, Mechanical Tubing, Rings, and Stock for Forgings, Rings, or Heading 15Cr - 25.5Ni - 1.2Mo - 2.1Ti - 0.006B - 0.30V (A286) Consumable Electrode Remelted 1800 °F (982 °C) Solution and Precipitation Heat Treated - UNS S66286
SAE AMS5853	-	Steel, Corrosion- and Heat-Resistant, Bars and Wire 15Cr - 25.5Ni - 1.2Mo - 2.1Ti - 0.006B - 0.30V Consumable Electrode Melted 1800 °F (982 °C) Solution Heat Treated and Work-Strengthened 160 ksi (1103 MPa) Tensile Strength Capability - UNS S66286
SAE AMS5962	-	Nickel Alloy, Corrosion and Heat-Resistant, Round Bars and Wire 52.5Ni - 19Cr - 3.0Mo - 5.1Cb - 0.90Ti - 0.50Al - 18Fe Consumable Electrode or Vacuum Induction Melted 1775 °F (968 °C) Solution Treated and Work Strengthened Precipitation Hardenable - UNS N07718
SAE AMS-QQ-P-416	-	Plating, Cadmium (Electrodeposited)
SAE AMS-H-81200	-	Heat Treatment of Titanium and Titanium Alloys
SAE J404	-	Chemical Compositions of SAE Alloy Steels
SAE J405	-	Chemical Compositions of SAE Wrought Stainless Steels

(Copies of these documents are available online at www.sae.org.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 Material and specific requirements. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.1.1 Alloy steel. Alloy steel screws shall be manufactured from Type 4037 (UNS G40370), Type 4137 (G41370), Type 4140 (G41400), Type 4142 (G41420), Type 4145 (G41450), Type 4340 (G43400), or Type 8740 (G87400) in accordance with the chemical compositions specified in SAE J404. Alloys with a minimum boron requirement are prohibited. Alloy steels shall be heat treated

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in accordance with SAE AMS2759/1, as required, to meet the mechanical properties specified in Table VIII. When specified in the contract (see 6.2), highly banded microstructures as shown in ASTM E1268 Appendix A, Figures A1.4 through A1.16, are not permitted, except as allowed by 4.3.6.

3.1.2 Corrosion-resistant steel (austenitic). Austenitic corrosion-resistant steel screws shall be manufactured from Type 302 (UNS S30200), Type 302HQ (UNS30430), Type 304 (UNS S30400), Type 304L (UNS S30403), Type 305 (UNS S30500), Type 316 (UNS S31600), Type 316L (UNS S31603) or Type 384 (UNS S38400) in accordance with the chemical compositions specified in SAE J405 and the mechanical properties as specified in Table VIII. Heat treatment and condition shall be in accordance with ASTM A493.

3.1.3 Heat and corrosion-resistant steel. Heat and corrosion-resistant steel screws shall be manufactured from A286 Iron Base Super alloy (UNS S66286) in accordance with SAE AMS5726, SAE AMS5731, SAE AMS5732 or SAE AMS5853. Cold work and precipitation hardening may be used to meet the mechanical properties specified in Table VIII. Heat and corrosion-resistant steel A286 shall be heat treated in accordance with SAE AMS2759/3.

3.1.4 Grade 23 titanium. Grade 23 titanium screws shall be manufactured from 6AL-4V ELI (UNS R56407) in accordance with ASTM B348/B348M. The metal used in the production of the bar, rod, or wire shall be produced by multiple melting using consumable electrode practice. The last melting cycle shall be under vacuum. Barstock used to make these fasteners shall be ultrasonic inspected in accordance with SAE AMS2631. Barstock 2.50 inches in diameter or smaller shall meet the requirements of SAE AMS2631 Acceptance Class A1. Barstock used to produce these fasteners shall be given a recrystallization anneal in accordance with SAE AMS-H-81200. Screws shall be hot worked or cold finished to meet the mechanical property requirements specified in Table VIII.

3.1.5 Grade 5 titanium. Grade 5 titanium screws shall be manufactured from 6AL-4V alloy (UNS R56400) in accordance with SAE AMS4928 or SAE AMS4967, heat treated in accordance with SAE AMS-H-81200 to meet the mechanical property requirements specified in Table VIII.

3.1.6 Nickel alloy 625. Nickel Alloy 625 screws shall be manufactured from nickel alloy 625 (UNS N06625) in accordance with ASTM B446, Grade 1. The metal used in the production of the bar, rod, or wire used to make Grade 625 fasteners shall be refined using the electroslag remelting process (ESR) or the vacuum arc remelting process (VAR). Fasteners shall be annealed after all manufacturing processes which can induce residual stresses are completed, and meet the mechanical property requirements specified in Table VIII.

3.1.7 Nickel alloy 718 – Aerospace. Nickel Alloy 718 screws shall be manufactured from nickel alloy 718 (UNS N07718) in accordance with SAE AMS5662 or SAE AMS5962, age hardened and cold reduced when necessary by rolling or drawing to meet the mechanical property requirements specified in Table VIII. Processing shall be in accordance with the pyrometry requirements of SAE AMS2750. Finished screw surfaces shall be acid cleaned unless otherwise specified and free from oxides and metal contaminants. This nickel alloy 718 is intended primarily for aerospace use. Navy shipboard applications shall instead use nickel alloy 718 - Navy per paragraph 3.1.8.

3.1.8 Nickel alloy 718 – Navy shipboard. Nickel Alloy 718 screws for Navy shipboard applications shall be manufactured from nickel alloy 718 (UNS N07718) per API Standard 6ACRA Material Designations 120K, 140K, or 150K.

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3.1.9 Nickel copper alloy. Nickel copper alloy screws shall be manufactured from Ni 400 series nickel copper alloy, (UNS N04400 or UNS N04405) in accordance with QQ-N-281 Class A or Class B, hot finished or cold drawn, work-hardened to meet the property requirements specified in Table VIII.

3.1.10 Nickel copper aluminum alloy. Nickel copper aluminum alloy screws shall be manufactured from nickel (Ni 500) series alloy (UNS N05500) in accordance with QQ-N-286, annealed and age hardened to meet the mechanical properties specified in Table VIII. Ni-Cu-Al material having been headed or roll threaded shall not be age hardened unless it has been solution annealed subsequent to the heading and threading operations. Threads formed after the final age hardening heat treatment shall be cut or ground only.

3.1.11 Naval brass. Naval brass screws shall be manufactured from copper alloy UNS C46400, temper designation code O50 to H02 (light anneal to half-hard) in accordance with ASTM B21/B21M to meet the mechanical properties specified in Table VIII.

3.1.12 Silicon bronze. Silicon bronze screws shall be manufactured from copper alloy UNS C65100 or UNS C65500, temper designation code H01 quarter-hard through H04 hard material in accordance with ASTM B98/B98M or ASTM B99/B99M to meet the properties specified in Table VIII.

3.1.13 Phosphor bronze. Phosphor bronze screws shall be manufactured from copper alloy UNS C51000 temper designation code H04 hard material in accordance with ASTM B139/B139M to meet the properties specified in Table VIII.

3.1.14 Nickel aluminum bronze. Nickel aluminum bronze screws shall be manufactured from copper alloy UNS C63200 material, temper designation code TQ50, quench hardened and temper annealed, in accordance with ASTM B150/B150M, to meet the properties specified in Table VIII.

3.1.15 Manganese bronze. Manganese bronze screws shall be manufactured from copper alloy UNS C67500, temper designation code O60 soft anneal, in accordance with ASTM B138/B138M, to meet the properties specified in Table VIII.

3.2 Mechanical properties. Screws shall meet the mechanical properties specified in Table VIII, as applicable.

3.2.1 DELETED.

3.2.2 DELETED.

3.2.3 DELETED.

3.2.4 DELETED.

3.2.5 DELETED.

3.2.6 All screws. Screws with hardness values up to 1 point below the specified minimum requirement in Table VIII may be accepted provided they pass all of the other requirements. Fasteners exceeding the maximum required hardness shall not be offered for acceptance.

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3.3 Protective coating or surface treatment. Screws shall be furnished uncoated or with a protective coating or surface treatment as specified herein (see 6.2 and 6.3).

3.3.1 Corrosion-resistant steel (austenitic). Corrosion-resistant steel screws shall be cleaned, descaled and passivated in accordance with ASTM A380/A380M, Code F, or SAE AMS2700, Method 1, Type 6, Type 7 or Type 8, or Method 2.

3.3.2 Heat and corrosion-resistant steel. Heat and corrosion-resistant steel screws shall be passivated in accordance with SAE AMS2700, Method 1, Type 8, or Method 2, no post treatment.

3.3.3 Cadmium plating (Obsolete for New Design, see 6.3.2). When specified, screws shall be cadmium plated /1 by electrodepositing in accordance with SAE AMS-QQ-P-416, Type II, Class 2 (Grade B preferred). When cadmium plating is specified for screws of 48 pitch or finer, plating thickness may be reduced below .0003 inch (8µm).

/1 Cadmium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. Zinc-nickel plating per section 3.3.5 may be used in lieu of cadmium with design authority approval.

3.3.4 Zinc plating. When specified, alloy steel screws shall be zinc plated with hexavalent chromium-free conversion coating by electrodepositing in accordance with ASTM F1941/F1941M, Class Fe/Zn 8, with CNS, DNS or ENS conversion coating. For zinc plating in accordance with ASTM F1941/F1941M, the fasteners shall receive a 23 hour bake at 375 °F ± 25 °F initiated within one hour after removal from the plating bath for hydrogen embrittlement relief treatment. When zinc plating is specified for screws of 48 pitch or finer, plating thickness may be reduced below .0003 inch (8 µm).

3.3.5 Zinc-Nickel plating. When specified, alloy steel screws shall be zinc-nickel plated in accordance with MIL-PRF-32647, Type K16 or K20, or SAE AMS2461, Type II, Grade B, Class 2. For the latter specification, the color shall be light to dark gray, but color shall not be a cause for rejection.

3.3.6 Phosphate coating. When specified, alloy steel screws shall be phosphate coated in accordance with MIL-DTL-16232, Type Z, Class 2; chromic acid rinse prohibited.

3.3.7 Black chemical finish. When specified, black oxide coatings shall be applied according to the following:

3.3.7.1 Alloy steel. When specified, alloy steel screws shall have a black chemical finish in accordance with MIL-DTL-13924, Class 1, or MIL-DTL-16232, Type M, Class 4.

3.3.7.2 Corrosion-resistant steel. When specified, corrosion-resistant steel screws shall have a black chemical finish in accordance with MIL-DTL-13924, Class 2 or 4 not utilizing hexavalent chromium, meeting the same performance requirements, or any other method that produces equivalent performance.

3.3.7.3 Heat and corrosion-resistant steel. When specified, heat and corrosion-resistant steel screws shall have a black chemical finish in accordance with MIL-DTL-13924, Class 3 or 4 not utilizing hexavalent chromium, meeting the same performance requirements, or any other method that produces equivalent performance.

3.3.7.4 Copper alloy. When specified, copper alloy screws shall have a black chemical finish in accordance with MIL-DTL-495 or an equivalent method not utilizing hexavalent chromium.

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3.3.8 Titanium alloy. Titanium screws shall have anodic treatment in accordance with SAE AMS2488, Type II modified as provided below. Unless an inorganic coating is specified in the part standard, a solid film lubricant to SAE AS1701 Class I shall be applied. When an inorganic coating is specified, a solid film lubricant to SAE AS1701 Class V or VI shall be applied. The anodic treatment shall be modified as follows:

a) The anodizing bath shall be PH-13 [alkaline] per SAE AMS2488.
b) The final anodize coating shall not cause any measurable dimensional change using standard fastener measuring equipment.
c) Confirmatory preproduction tests of SAE AMS2488 are not required.

3.4 Surface texture. For alloy steel screws of sizes up to and including .625 inch, and having nominal lengths equal to or less than 8 times the basic screw diameter, the surface roughness of the screws before plating shall not exceed 63 microinches RA on the fillet and head bearing surfaces, nor exceed 32 microinches RA on the threads. For larger sizes, longer lengths, and corrosion-resistant steel screws, the surface roughness of the screws before plating shall not exceed 125 microinches RA on the body, fillet and head bearing surfaces. Normally, it shall be sufficient to ascertain these surfaces on screws have the equivalent of a smooth machined finish by visual comparison with known surface standards. However, when it is deemed necessary to measure these surfaces with commercially available equipment, roughness measurements shall be taken axially on the body and fillet surfaces, and circumferentially on the bearing surface in accordance with ASME B46.1.

3.5 Design.

3.5.1 Dimensions. Screws shall conform to the dimensions, definitions and tolerances specified in ASME B18.3. Exceptions shall be noted per 6.2.

3.5.2 Cylindrical heads (drilled). When specified (see 6.2), screws shall be cross-drilled in accordance with Figure 2 and Table I. The alignment plug specified in Table I shall pass completely through the head without deflection. On nominal sizes .164 inch and above, the drilled holes shall not break through the corners of the hexagon socket. Nominal sizes .112 inch and .138 inch shall have two drilled holes spaced 180°; sizes .164 inch and above shall have six drilled holes spaced 60°. Edges of holes in the sockets may contain burrs; however, the socket shall accept a key in accordance with ASME B18.3.

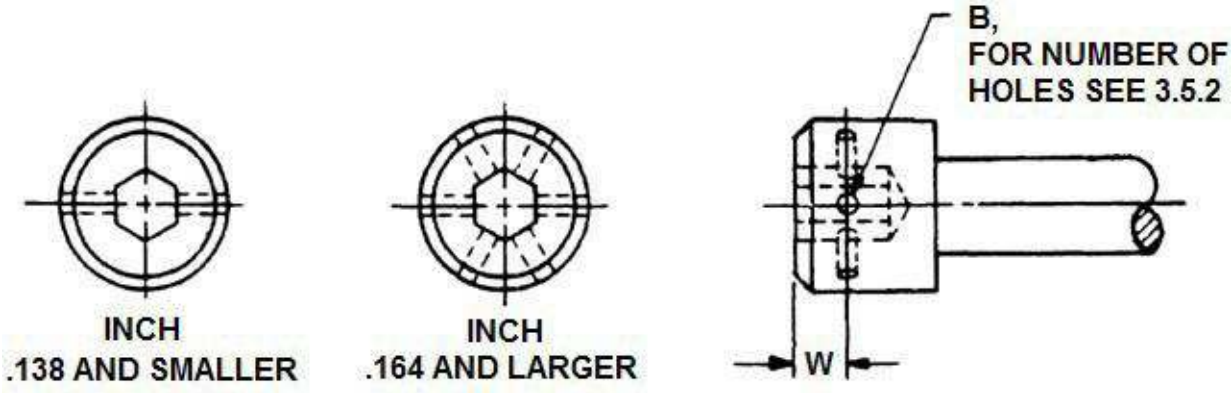


Figure 2. Cylindrical head drilling configurations.

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3.5.3 Broaching chips. Broached sockets of screws nominal size .112 or larger shall have all loose chips removed. Remaining chips shall be firmly attached.

3.5.4 Threads.

3.5.4.1 Method of manufacture of threads on screws (except Ni-Cu-Al). Unless otherwise specified by the procuring agency or contract, rolled threads are mandatory for all sizes, except rolled, cut or ground threads are acceptable for finished nominal diameters greater than one inch provided all other requirements herein are met.

3.5.4.2 Method of manufacture of threads on nickel-copper-aluminum fasteners (Ni-Cu-Al, K-500). Ni-Cu-Al material having been headed or roll threaded shall not be age hardened unless it has been solution annealed subsequent to the heading and threading operations. Threads formed after the final age hardening heat treatment shall be cut or ground only.

Table I. Dimensions for drilled cylindrical head screws.

NOMINAL SIZE	W TOP OF HEAD TO CENTER OF HOLE		B DRILLED HOLE DIAMETER		HOLE ALIGNMENT PLUG DIAMETER
	MAX	MIN	MAX	MIN	
.1120	.040	.026	.039	.033	.025
.1380	.050	.035	.039	.033	.025
.1640	.060	.040	.050	.044	.030
.1900	.065	.045	.050	.044	.030
.2500	.085	.065	.050	.044	.030
.3125	.104	.084	.050	.044	.030
.3750	.123	.103	.067	.061	.051
.4375	.141	.121	.067	.061	.051
.5000	.160	.140	.067	.061	.051
.6250	.198	.178	.067	.061	.051
.7500	.235	.215	.097	.091	.081
.8750	.273	.253	.097	.091	.081
1.0000	.310	.290	.097	.091	.081
1.1250	.348	.328	.127	.119	.104
1.2500	.385	.365	.127	.119	.104
1.3750	.423	.403	.127	.119	.104
1.5000	.460	.440	.127	.119	.104

3.5.4.3 Thread series and class. Unless otherwise specified (see 6.2), threads shall be UNC, UNRC, UNF, UNRF or 8UN series: Typical Class 3A for nominal diameter sizes .060 through 1.000 inch inclusive and class 2A for nominal diameter sizes over 1.000 inch and larger in accordance with ASME B1.1. Older designs using FED-STD-H28/2 are still acceptable. When applying coating to Class 2A, the maximum material limit of Class 3A shall be met after application.

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3.5.5 Source identification mark. All screws with nominal diameters .1900 and larger shall be permanently marked to identify the source accepting responsibility for the screws meeting all requirements specified herein, including coatings. The marking shall be a source identifying symbol for a manufacturer in accordance with MIL-HDBK-57 or a private label distributor's symbol as applicable, and shall not be on a bearing surface.

3.5.6 Material identification mark. Screws with nominal size .2500 and larger shall be permanently marked to identify the material specified in 3.1. The marking shall be raised or depressed at the option of the manufacturer and located on the top of the screw head in such a manner the marking is visible after installation and any applied coating, unless otherwise specified on the applicable specification sheet (see Table II).

TABLE II. Material identification mark.

Material	Marking
Alloy Steel /1	
AISI 302	302
AISI 302HQ /1	
AISI 304	B8 or 304
AISI 304L	304L
AISI 305	B8P
AISI 316	B8M or 316
AISI 384	384
AISI 316L	316L
Heat and Corrosion-Resistant Steel	A286 or N
Manganese Bronze	675/2
Nickel Aluminum Bronze	632/2
Phosphor Bronze	510/2
Silicon Bronze 651	651/2
Silicon Bronze 655	655
Naval Brass	464/2
Nickel Copper	NICU or NC
Nickel Copper Aluminum	.K. or ·K·
Nickel alloy 625	625/2
Nickel alloy 718 (Aerospace)	718
Nickel alloy 718 (Navy shipboard) Material Designation 120K	718N1
Nickel alloy 718 (Navy shipboard) Material Designation 140K	718N2
Nickel alloy 718 (Navy shipboard) Material Designation 150K	718N3
Titanium Grade 23	T23
Titanium Grade 5	T5

/1 Alloy steel and 302HQ do not require material marking.
/2 Use of parts made to a prior revision are acceptable until stock is depleted.

3.6 Carburization and decarburization limits for alloy steel screws. For alloy steel screws, the extent of carburization and decarburization shall be in accordance with the requirements of ASTM A574 and ASTM F835, as applicable (see 4.3.2).

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3.7 Discontinuities.

3.7.1 Definitions.

3.7.1.1 Seam. A seam is a longitudinal, elongated discontinuity in the metal caused by a defect which has been largely closed by rolling or forging mechanically, but not united. See Figure 3.

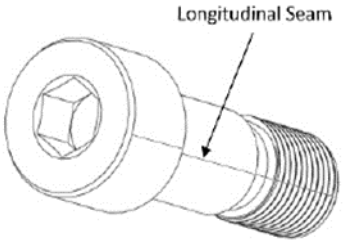


Figure 3. Seam in a cylindrical socket head cap screw.

3.7.1.2 Lap. A lap is a surface defect appearing as a seam caused by folding over metal and then rolling or forging, but not welding them into the surface.

3.7.1.3 Shank. The portion of the part approximately at the nominal diameter of the part, including both the unthreaded body of the part – if any – as well as the threaded portion of the part.

3.7.1.4 Head. The portion of the part materially larger than the nominal diameter of the shank, herein characterized as either a “cylindrical head”, “button head” or “flat head”.

3.7.1.5 Basic thread height. The basic thread height is defined as .8660254 divided by the number of threads per inch on the fastener. For instance, a quarter-inch diameter screw with 20 threads per inch would have a basic thread height of approximately .0433 inches.

3.7.2 Thread discontinuities. Acceptable discontinuities in the threads, and the limits associated therewith, depend on the nature of the discontinuity and its location, as described below. Discontinuities other than those specifically deemed acceptable herein are not allowed. All discontinuities are to be measured perpendicular to the indicated surfaces.

3.7.2.1 Longitudinal seams in threads. Threads may contain longitudinal seams and longitudinal seam-related laps at any point, to the depth limits in 3.7.4.

3.7.2.2 Lateral discontinuities in threads. Threads shall have no lateral discontinuities at the root or on the flanks below the pitch diameter line as shown in Figure 4. Laps and seams are permissible at the crest and on thread flanks above the pitch diameter line to a depth of 25 percent of the basic thread height, as shown in Figure 5. Crests may be slightly deformed, as shown in Figure 6. Other than these permissible discontinuities, no other defects are allowed in the threads.

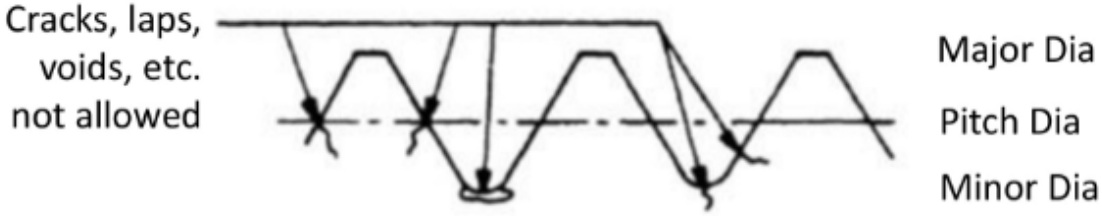


Figure 4. Disallowed lateral thread discontinuities.

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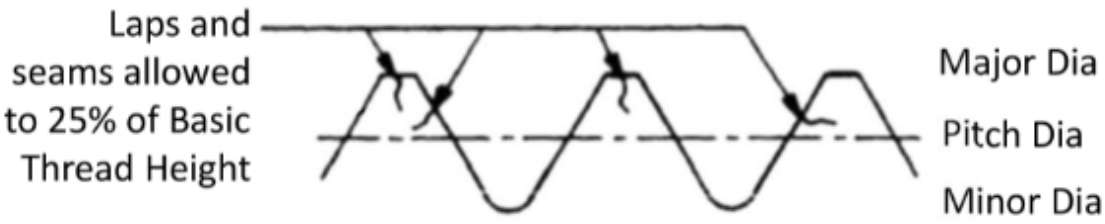


Figure 5. Permissible lateral thread discontinuities.

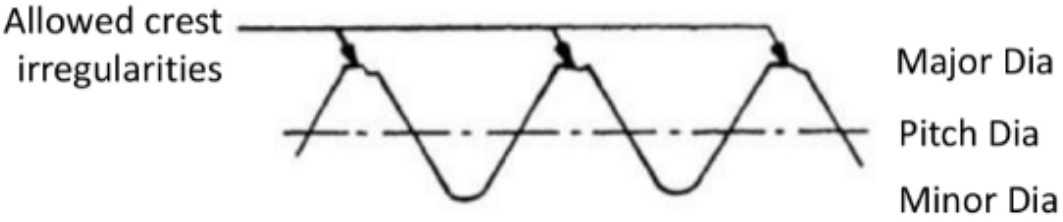


Figure 6. Permissible crest irregularities.

3.7.2.3 Thread nicks. Thread nick acceptability shall be determined in accordance with ASTM F788/F788M-2020 version, section 6.5.1.4 and its Table 1. Thread nicks shall be minimized per 6.5.

3.7.3 All other discontinuities. Discontinuities in locations other than in the threads, including but not limited to the socket, top of the head, side of the head, bearing surface, fillet, and any unthreaded portion of the shank are acceptable to the depth limits in 3.7.4.

3.7.4 Discontinuity depth limits. For specific allowed discontinuities other than those in section 3.7.2.2, the maximum depth measured perpendicular to the indicated surface is .03 times the nominal diameter (rounded to four decimal places) of the part. For quick reference, Table III below provides those limits for common nominal diameters.

Table III. Discontinuity limits.

Nominal Diameter	Max Discontinuity Depth
.0600	.0018
.0730	.0022
.0860	.0026
.0990	.0030
.1120	.0034
.1380	.0041
.1640	.0049
.1900	.0057
.2500	.0075
.3125	.0094
.3750	.0113

Nominal Diameter	Max Discontinuity Depth
.4375	.0131
.5000	.0150
.5625	.0169
.6250	.0188
.7500	.0225
.8750	.0263
1.0000	.0300
1.1250	.0338
1.2500	.0375
1.3750	.0413
1.5000	.0450

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4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use their own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the customer. The customer reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lot. A lot shall consist of all screws of the same type, made from the same heat of material, heat treatment, protective coating, surface finish, size, length, and threads produced and manufactured at the same time by the same manufacturer's facility and offered for acceptance at the same time. For production efficiencies, a lot of parts may be produced from a single component lot by the original manufacturer of the component lot as long as the mechanical and chemical properties of the component lot remain unchanged in subsequent production (see 6.8). In the event the component lot can provide test data (e.g. wedge tensile strength, tensile strength, yield, elongation or double shear), this test data may be used to certify the finished lot.

4.2.2 Sampling for examination. A random sample of screws shall be taken from each lot in accordance with ANSI/ASQ Z1.4, using single sampling plans Table I and Table II-A, General Inspection level II. AQL's per Table V herein. A C=0 sampling plan which achieves the same AQL's per ASQ H1607, Table 1 may be utilized in lieu of ANSI/ASQ Z1.4. For small lot procurement, see 4.2.9.

4.2.3 DELETED.

4.2.4 DELETED.

4.2.5 Sampling for mechanical properties and metallographic tests. A random sample of screws shall be taken from each lot in accordance with ANSI/ASQ Z1.4, using single sampling plans Table I and Table II-A, Inspection level S-1, & AQL 4.0, unless otherwise specified by the contract for destructive testing (see 6.2). For small lot procurement, see 4.2.9.

4.2.6 Sampling for protective coating or surface treatment tests. Sampling depends on the type of test being conducted and the coating or surface treatment utilized.

4.2.6.1 Corrosion resistance testing. Corrosion resistance testing is only required when it is required by the relevant coating/surface treatment standard, part standard, or when specifically required by the contract (see 6.2). When required, the number of samples shall be in accordance with Table IV, unless a greater number is required by the relevant coating/surface treatment standard or part standard.

4.2.6.2 Hydrogen embrittlement testing. Hydrogen embrittlement testing is required for all lots of alloy parts plated by electrodeposition and with hardness HRC 39 or above (unless a lower hardness threshold is set by the coating/part standard). In addition to in-process controls required by the relevant coating standard (if any), a minimum of four (4) samples shall be randomly drawn from the lot for testing unless a higher number is required by the coating standard or part standard.

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Table IV. Number of samples per lot size.

Lot Size	Number of Samples	Accept	Reject
Less than 151	1	0	1
151 to 10,000	2	0	1
10,001 to 35,000	3	0	1
Greater than 35,000	4	0	1

Table V. Classification of defects.

CATEGORIES	DEFECTS	INSPECTION METHOD
CRITICAL	AQL = 1.00	
1	Discontinuities (see 3.7)	CIE 1/
2	Source identification mark (see 3.5.5)	Visual
3	Material identification mark (except alloy steel and 302HQ) (see 3.5.6)	Visual
Major	AQL = 2.50	
101	Socket dimensions (see 3.5.1)	CIE
102	Threads (see 3.5.4)	CIE
Minor	AQL = 4.00	
201	Protective coating or surface treatment (see 3.3)	Visual/CIE
202	Surface texture (see 3.4)	Visual/CIE
203	Body diameter (cylindrical and flat countersunk heads) (see 3.5.1)	CIE
204	Head diameter (see 3.5.1)	CIE
205	Head height (see 3.5.1)	CIE
206	Head chamfer (see 3.5.1)	CIE
207	Bearing surface (see 3.5.1)	CIE
208	Edge of head radius or chamfer (cylindrical heads) (see 3.5.1)	CIE
209	Concentricity (see 3.5.1)	CIE
210	Under head fillet (see 3.5.1)	CIE
211	Screw length (see 3.5.1)	CIE
212	Thread length (see 3 5 .1)	CIE
213	Grip gaging length (cylindrical and flat countersunk heads) (see 3.5.1)	CIE
214	Screw point chamfer (see 3.5.1)	CIE
215	Flushness tolerance (flat countersunk heads) (see 3.5.1)	CIE
216	Drilled hole dimensions (cylindrical heads) (see 3.5.2)	CIE
217	Broaching chips (see 3.5.3)	Visual/ CIE

/1 "CIE" = Commercial Inspection Equipment

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4.2.7 Sampling for chemical analysis. A random sample of screws shall be taken from each lot in accordance with ANSI/ASQ Z1.4, using single sampling plans Table I and Table II-A, Inspection level S-1, & AQL 1.0. Use of mill certification for chemical analysis shall be acceptable (see 4.3.4) unless otherwise specified by the contract (see 6.2). For small lot procurement, see 4.2.9.

4.2.8 Sampling for discontinuities (non-destructive testing). A random sample of screws shall be taken from each lot in accordance with ANSI/ASQ Z1.4, using single sampling plans Table I and Table II-A, Inspection general level II, & AQL 1.0. For small lot procurement, see 4.2.9.

4.2.9 Sampling for small lots. For small lot procurement of less than 51 screws, sampling and acceptance/rejection criteria for mechanical properties, metallographic, chemical analysis, and discontinuity tests shall be as follows:

TABLE VI. Sampling for small lots.

		Number of Defectives	
Lot size	Sample size	Accept	Reject
2 to 10	1	0	1
11 to 50	2	0	1

4.2.10 DELETED.

4.2.11 Examination.

4.2.11.1 Visual and dimensional examination. Each screw taken as specified in 4.2.2 shall be examined to verify conformance with this specification. Examination shall be conducted in accordance with Table V (in accordance with ANSI/ASQ Z1.4, using single sampling plans Table I and Table II-A, Inspection general level II or a C=0 sampling plan which achieves the same AQL's per ASQ H1607, Table 1 may be utilized in lieu of ANSI/ASQ Z1.4) and ASME B18.3. Any screw in the sample containing one or more defects shall be rejected, and if the number of defective screws in any sample exceeds the acceptance number for its category, the lot represented by the sample shall be rejected. Alternately, at the procuring agency's discretion, 100% inspection in accordance with ASME B18.18 may be permitted.

4.2.11.2 Thread inspection. Screw threads shall be inspected in accordance with ASME B1.3, System 22.

4.3 Test methods.

4.3.1 Mechanical property test. Mechanical tests shall be performed on each lot of fasteners per Figure 7 based on overall size, head type, nominal diameter, and length of the part. Screws shall be tension tested full size, whenever possible. When the actual fastener cannot be tested for conformance to requirements, alternate methods may be used as provided in subparagraphs below. However, hardness shall/1 be tested on finished parts, even when alternative methods are used for other requirements. Acceptance criteria for each test is provided in Table VIII. Testing is not required for any mechanical test attribute not having acceptance criteria for its attribute.

/1 In rare cases, when small parts are being machined from hardened bar stock and no additional thermal processing will be used, all mechanical properties below may be confirmed on bar samples machined from the full-diameter bar stock.

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4.3.1.1 Hardness test. Samples selected as specified in 4.2.5 and in accordance with 4.3.1 shall be subjected to a hardness test conducted in accordance with ASTM F606/F606M at room temperature on a clean, flat surface, capable of sustaining the load imposed by the hardness tester. Acceptance criteria is per Table VIII.

4.3.1.2 Wedge tensile test. When wedge tensile testing is required by Figure 7, samples selected as specified in 4.2.5 shall be subjected to wedge tests conducted in accordance with ASTM F606/F606M, with wedge angles in accordance with Table VII. For large parts, testing as allowed per 4.3.1.4 may be substituted for wedge tensile testing. For parts less than .112" in diameter, perform axial tensile test per 4.3.1.3 in lieu of wedge tensile testing. Acceptance criteria is per Table VIII. Fasteners having been wedge tested shall not be submitted for acceptance.

4.3.1.3 Axial tensile test. When axial tensile testing is required by Figure 7, samples selected as specified in 4.2.5 shall be subjected to a full-size axial tension tests conducted in accordance with ASTM F606/F606M. For cylindrical socket head cap screws (head type I and VI per 1.2.1), wedge tensile tests in accordance with 4.3.1.2 may be substituted for axial tensile tests. For large parts, testing as allowed per 4.3.1.4 may be substituted for axial tensile testing. For flat countersunk and button head screws (head type IV and VIII per 1.2.1), the fracture may extend or spread into the head of the fastener. Acceptance criteria is per Table VIII. Fasteners having been axial tested shall not be submitted for acceptance.

4.3.1.4 Wedge/axial tensile testing of large parts. For testing of screws having tensile properties greater than can be measured on available test equipment, machined test specimens may be substituted for the full-sized part wedge or axial tensile mechanical tests required in Figure 7. The machined test specimens shall have the body turned down, concentric to the axis of the screw, to the largest practical gage dimension specified in ASTM F606/F606M. These specimens shall be tested for axial tensile strength, elongation and reduction of area in accordance with ASTM F606/F606M. Additionally, yield strength testing shall be performed on machined test specimens as provided in 4.3.1.5 when specified by contract.

4.3.1.5 Yield strength/proof load testing. When yield strength and/or proof load testing is required by the applicable contract and Figure 7, samples selected as specified in 4.2.5 shall be subjected to full-size tests conducted in accordance with 4.3.1.5.1 and/or 4.3.1.5.2, below. When full-size testing is not practical, machined samples prepared in accordance with ASTM E8/E8M, section 6.4 may be substituted for full-size parts. Acceptance criteria on machined samples remain unchanged.

4.3.1.5.1 Yield strength test. Yield strength testing shall be conducted using the offset method detailed in ASTM F606/F606M. Acceptance criteria is per Table VIII.

4.3.1.5.2 Proof load test. Proof load testing on Type I or Type VI cylindrical socket head cap screws shall be conducted in accordance with ASTM F606/F606M Method 1. Acceptance criteria is per Table VIII, note 5.

4.3.1.6 Testing for short parts. When part length is less than three times the nominal diameter (3D) or there is insufficient length for proper fit up in test equipment, mechanical properties shall be verified by one of the methods set forth in 4.3.1.6.1, 4.3.1.6.2 or 4.3.1.6.3. Such alternative testing shall be allowed only when the fastener length does not meet the

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minimum 3D length requirement or there is insufficient length for proper fit up in test equipment. Unless otherwise specified by the contract, the choice of method is up to the manufacturer.

4.3.1.6.1 Double shear. Four (4) test specimens shall be taken from the raw material used to produce the lot, thermally processed with the manufacturing lot being tested, and double shear tested in accordance with NASM1312-13. The test specimen, or a portion thereof, shall incorporate a cylindrical element with a length equal to, or greater than, two (2) times the cylindrical diameter of the test specimen. Acceptance criteria is per Table VIII.

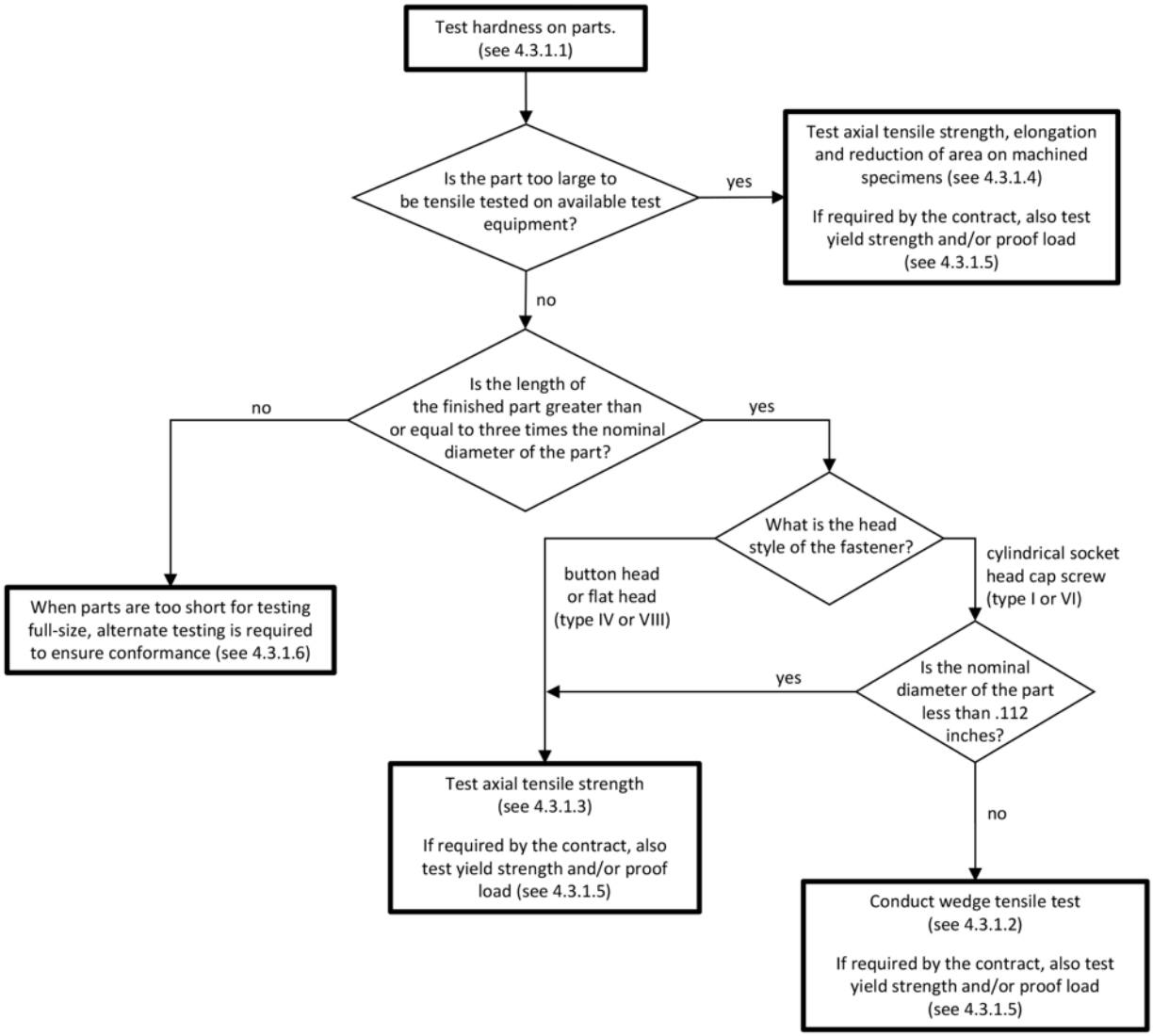


Figure 7. Mechanical testing requirements decision tree.

4.3.1.6.2 Machined specimens. Four (4) test specimens shall be machined from samples taken from the manufacturing lot being tested. The body shall be turned down, concentric to the axis of the screw, to the largest practical gage dimension specified in ASTM F606/F606M. These specimens shall be tested for axial tensile strength, elongation and reduction of area in accordance with ASTM F606/F606M. Acceptance criteria is per Table VIII. In addition, when specified by contract, yield strength testing may be performed in accordance with ASTM F606/F606M, section 3.6.1, with specimens prepared in accordance with ASTM E8/E8M, section 6.4.

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4.3.1.6.3 Longer test pieces. Four (4) longer fasteners may be utilized to verify lot mechanical properties. Gage length shall not be increased by machining threads into the shank. The longer test fasteners shall be made from the same heat of rod, bar or wire used for the production fasteners; have the same type and nominal diameter as the production fasteners; and shall be processed and heat treated with the production fasteners. When parts are being machined from hardened bar stock and no additional thermal treatments are to be used, longer test pieces may be formed by increasing the length and number of threads while maintaining the same nominal diameter and body length as the production fasteners. Acceptance criteria is per Table VIII. Special care shall be utilized to ensure ALL longer pieces are removed from the lot to avoid inclusion of foreign object detected/debris (FOD) when parts are shipped.

Table VII. Wedge angles for Type I and Type VI cylindrical socket head cap screws.

Material	Size	Wedge Angle /1	
		Body Lengths greater than 2D /2	Body Lengths 2D /2 or less or Threaded to the Head
Alloy Steel	.112-.50 incl.	10	6
	Over .50-under .875	8	6
	.875 incl. and larger	6	4
CRES and Heat and Corrosion-Resistant Steel	.112-.750 incl.	10	6
	Over .750	6	4
Nonferrous (except titanium)	.112-.750 incl.	10	6
	Over .750	6	4
Titanium	.112-.750 incl.	6	4
	Over .750	4	4

/1 Wedge angles listed are minimum values, in degrees.
/2 D is the nominal diameter of the part.

4.3.2 Metallographic test. Alloy steel samples selected as specified in 4.2.5 shall be subjected to a metallographic test conducted in accordance with ASTM A574 or ASTM F835 to determine conformance to carburization and decarburization limits (see 3.6).

4.3.3 Protective coating or surface treatment tests. Protective coating or surface treatment tests shall be conducted in accordance with the following requirements.

4.3.3.1 Corrosion resistance testing. Fastener samples selected per 4.2.6.1 shall be tested in accordance with the relevant part/coating standard.

4.3.3.2 Hydrogen embrittlement testing. Electroplated alloy steel screws shall be subjected to the hydrogen embrittlement relief requirement specified in the applicable electrodeposited plating or coating specification (see 3.3). Fastener samples selected per 4.2.6.2 shall be tested in accordance with the relevant coating/part standard. Following removal, samples shall be inspected per ASTM E1444/E1444M with acceptance criteria per section 3.7 above.

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4.3.4 Chemical Analysis. Samples selected (see 4.2.7) shall be subjected to a chemical analysis in accordance with ASTM E1282. When a mill certificate covering chemical analysis of utilized raw materials can be furnished, it shall be acceptable in lieu of the chemical analysis of the finished part required by 4.2.7 (see 6.2).

4.3.5 Discontinuity inspection test. For samples selected (see 4.2.8), non-ferromagnetic screws shall be subjected to liquid penetrant testing in accordance with ASTM E1417/E1417M, Type I or Type II, Method A, B or D, and ferromagnetic screws shall be subjected to magnetic particle testing in accordance with ASTM E1444/E1444M inspection methods as applicable to meet the requirements (see 3.7), except etching shall not be used for alloy steel screws intended for anything other than testing. If indications are identified, the samples shall be metallurgically examined within a range from 15X to 200X magnification to determine the extent and depth of the discontinuities. NAVSEA Technical Publication T9074-AS-GIB-010/271 may be substituted for ASTM E1417/E1417M for liquid penetrant testing or E1444/E1444M for magnetic particle testing. QQ-N-286 nickel copper aluminum alloy screws shall be inspected in accordance with T9074-AS-GIB-010/271.

4.3.6 Banded structure inspection. When banded structure inspection is specified in the contract (see 6.2), the sample shall be inspected in accordance with ASTM E1268 for banded microstructure. If a highly banded structure is observed, perform micro-hardness readings in accordance with ASTM E1268 on the most predominate bands. The test load shall be 500 gf, unless extremely thin bands are observed not permitting an accurate 500 gf test, in which case it is permissible to reduce the test load to as low as 50 gf to obtain an accurate reading. Individual readings shall not exceed 470 HK (or converted). One test shall be conducted per heat.

4.3.7 Rounding of test results. If rounding is necessary, results of mechanical and chemical analysis tests shall be rounded in accordance with the rounding method of ASTM E29. Results of the chemical analysis shall be rounded to the number of significant digits of the applicable test requirement. Wedge and/or axial tensile strength test values of up to 50,000 psi shall be rounded to the nearest 100 psi; test values over 50,000 psi and less than 100,000 psi shall be rounded to the nearest 500 psi; and test values over 100,000 psi shall be rounded to the nearest 1,000 psi. Elongation and reduction of area test values of up to 10 percent shall be rounded to the nearest 0.5 percent and test values over 10 percent shall be rounded to the nearest 1 percent.

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TABLE VIII. Mechanical properties of fasteners (see 3.1 for Materials).

Material	Head Type	Nominal Diameter	Hardness	Minimum Ultimate Tensile Strength PSI /3	Minimum Yield Strength PSI /4	Minimum Shear Strength (F_{SU}) PSI /1 /9	Minimum Elongation in 4D	Minimum Reduction of Area
Alloy Steel – Full Size /5	Type I or VI/ cylindrical socket head cap screw	≤ .500 inch	39 to 44 HRC /2	180,000	155,000			
		> .500 inch	37 to 44 HRC /2	170,000	150,000			
Alloy Steel – Full Size	Type IV or VIII/ countersunk, button head	≤ .500 inch	39 to 44 HRC /2	144,000	124,000			
		> .500 inch	37 to 44 HRC /2	136,000	120,000			
Alloy Steel – Machined Test Specimens	all	from parts ≤ .500 inch	39 to 44 HRC /2	180,000	155,000	108,000	12%	33%
		from parts > .500 inch	37 to 44 HRC /2	170,000	150,000	102,000	12%	33%
Corrosion-Resistant Steel (austenitic) – Full Size	Type I or VI/ cylindrical socket head cap screw	≤ .625 inch	80 HRB min	80,000	30,000			
	Type I or VI/ cylindrical socket head cap screw	> .625 inch	74 HRB min	70,000	26,000			
Corrosion-Resistant Steel (austenitic) – Full Size	Type IV or VIII/ countersunk, button head	≤ .625 inch	80 HRB min	64,000	24,000			
	Type IV or VIII/ countersunk, button head	> .625 inch	74 HRB min	56,000	20,800			
Corrosion-Resistant Steel (austenitic) – Machined Test Specimens	all	from parts ≤ .625 inch	80 HRB min	80,000	30,000	60,000	10%	30%
	all	from parts > .625 inch	74 HRB min	70,000	26,000	52,500	20%	30%
Heat and Corrosion-Resistant Steel – Full Size	Type I or VI/ cylindrical socket head cap screw	all	33 to 42 HRC	160,000	120,000			
Heat and Corrosion-Resistant Steel – Full Size	Type IV or VIII/ countersunk, button head	all	33 to 42 HRC	128,000	96,000			
Heat and Corrosion-Resistant Steel – Machined Test Specimens	all	all	33 to 42 HRC	160,000	120,000	95,000 /6	12%	18%

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TABLE VIII. Mechanical properties of fasteners (see 3.1 for Materials) - Continued.

Material	Head Type	Nominal Diameter	Hardness	Minimum Ultimate Tensile Strength PSI /3	Minimum Yield Strength PSI /4	Minimum Shear Strength (F_{SU}) PSI /1 /9	Minimum Elongation in 4D	Minimum Reduction of Area
Titanium Grade 23 – Full Size	Type I or VI/ cylindrical socket head cap screw	all	25 to 36 HRC	120,000 min 165,000 max	110,000			
Titanium Grade 23 – Full Size	Type IV or VIII/ countersunk, button head	all	25 to 36 HRC	96,000 min 165,000 max	88,000			
Titanium Grade 23 – Machined Test Specimens	all	all	25 to 36 HRC	120,000min 165,000 max	110,000	70,000	12%	
Titanium Grade 5 – Full Size	Type I or VI/ cylindrical socket head cap screw	all		160,000 /7	150,000 /10			
Titanium Grade 5 – Full Size	Type IV or VIII/ countersunk, button head	all		128,000	120,000 /10			
Titanium Grade 5 – Machined Test Specimens	all	all		160,000 /7	150,000 /10	95,000 /7		
Nickel Alloy 625 – Full Size	Type I or VI/ cylindrical socket head cap screw	all	85 HRB min 35 HRC max	120,000	60,000			
Nickel Alloy 625 – Full Size	Type IV or VIII/ countersunk, button head	all	85 HRB min 35 HRC max	96,000	48,000			
Nickel Alloy 625 – Machined Test Specimens	all	all	85 HRB min 35 HRC max	120,000	60,000	72,000	30%	
Nickel Alloy 718 Aerospace – Full Size/8	Type I or VI/ cylindrical socket head cap screw	all		220,000 min 250,000 max	200,000 /11			
Nickel Alloy 718 Aerospace – Full Size/8	Type IV or VIII/ countersunk, button head	all		176,000 min 250,000 max	160,000 /11			
Nickel Alloy 718 Aerospace – Machined Test Specimens/8	all	all		220,000 min 250,000 max	200,000 /11	125,000	8%	15%
Nickel Alloy 718 Navy Shipboard Material Designation 120K – Full Size	Type I or VI/ cylindrical socket head cap screw	all		150,000 min	120,000 min 140,000 max			
Nickel Alloy 718 Navy Shipboard Material Designation 120K – Full Size	Type IV or VIII/ countersunk, button head	all		120,000 min	96,000 min 112,000 max			

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TABLE VIII. Mechanical properties of fasteners (see 3.1 for Materials) - Continued.

Material	Head Type	Nominal Diameter	Hardness	Minimum Ultimate Tensile Strength PSI /3	Minimum Yield Strength PSI /4	Minimum Shear Strength (F_{SU}) PSI /1 /9	Minimum Elongation in 4D	Minimum Reduction of Area
Nickel Alloy 718 Navy Shipboard Material Designation 120K – Machined Test Specimens	all	all		150,000 min	120,000 min 140,000 max		20%	35%
Nickel Alloy 718 Navy Shipboard Material Designation 140K – Full Size	Type I or VI/ cylindrical socket head cap screw	all		165,000 min	140,000 min 150,000 max			
Nickel Alloy 718 Navy Shipboard Material Designation 140K – Full Size	Type IV or VIII/ countersunk, button head	all		132,000 min	112,000 min 120,000 max			
Nickel Alloy 718 Navy Shipboard Material Designation 140K – Machined Test Specimens	all	all		165,000 min	140,000 min 150,000 max		20%	35%
Nickel Alloy 718 Navy Shipboard Material Designation 150K – Full Size	Type I or VI/ cylindrical socket head cap screw	all		175,000 min	150,000 min 175,000 max			
Nickel Alloy 718 Navy Shipboard Material Designation 150K – Full Size	Type IV or VIII/ countersunk, button head	all		140,000 min	120,000 min 140,000 max			
Nickel Alloy 718 Navy Shipboard Material Designation 150K – Machined Test Specimens	all	all		175,000 min	150,000 min 175,000 max		20%	35%
Nickel Copper Aluminum Alloy – Full Size	Type I or VI/ cylindrical socket head cap screw	< 1.000 inch	24 to 35 HRC	130,000 min 180,000 max	90,000			
	Type I or VI/ cylindrical socket head cap screw	≥ 1.000 inch	24 to 35 HRC	130,000 min 180,000 max	85,000			
Nickel Copper Aluminum Alloy – Full Size	Type IV or VIII/ countersunk, button head	< 1.000 inch	24 to 35 HRC	104,000 min 180,000 max	72,000			
	Type IV or VIII/ countersunk, button head	≥ 1.000 inch	24 to 35 HRC	104,000 min 180,000 max	68,000			

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TABLE VIII. Mechanical properties of fasteners (see 3.1 for Materials) - Continued.

Material	Head Type	Nominal Diameter	Hardness	Minimum Ultimate Tensile Strength PSI /3	Minimum Yield Strength PSI /4	Minimum Shear Strength (F_{SU}) PSI /1 /9	Minimum Elongation in 4D	Minimum Reduction of Area
Nickel Copper Aluminum Alloy -- Machined Test Specimens	all	from parts < 1.000 inch	24 to 35 HRC	130,000	90,000	78,000	20%	
	all	from parts ≥ 1.000 inch	24 to 35 HRC	130,000	85,000	78,000	20%	
Nickel-Copper Alloy – Full Size	Type I or VI/ cylindrical socket head cap screw	all	75 HRB min 25 HRC max	80,000	40,000			
Nickel-Copper Alloy – Full Size	Type IV or VIII/ countersunk, button head	all	75 HRB min 25 HRC max	64,000	32,000			
Nickel-Copper Alloy – Machined Test Specimens	all	all	75 HRB min 25 HRC max	80,000	40,000	48,000	20%	
Naval Brass – Full Size	Type I or VI/ cylindrical socket head cap screw	all	55 to 75 HRB	50,000	27,000			
Naval Brass – Full Size	Type IV or VIII/ countersunk, button head	all	55 to 75 HRB	40,000	21,600			
Naval Brass – Machined Test Specimens	all	from parts ≤ .500 inch	55 to 75 HRB	50,000	27,000	30,000	22%	
	all	from parts > .500 inch	55 to 75 HRB	50,000	27,000	30,000	25%	
Silicon Bronze – Full Size	Type I or VI/ cylindrical socket head cap screw	all	60 to 95 HRB	60,000	40,000			
Silicon Bronze – Full Size	Type IV or VIII/ countersunk, button head	all	60 to 95 HRB	48,000	32,000			
Silicon Bronze – Machined Test Specimens	all	all	60 to 95 HRB	60,000	40,000	36,000	10%	
Phosphor Bronze Cap Screws – Full Size	Type I or VI/ cylindrical socket head cap screw	all	60 to 95 HRB	60,000 min 90,000 max	35,000			
Phosphor Bronze Cap Screws – Full Size	Type IV or VIII/ countersunk, button head	all	60 to 95 HRB	48,000 min 90,000 max	28,000			
Phosphor Bronze – Machined Test Specimens	all	all	60 to 95 HRB	60,000 min 90,000 max	35,000	36,000	15%	

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TABLE VIII. Mechanical properties of fasteners (see 3.1 for Materials) - Continued.

Material	Head Type	Nominal Diameter	Hardness	Minimum Ultimate Tensile Strength PSI /3	Minimum Yield Strength PSI /4	Minimum Shear Strength (F_{SU}) PSI /1 /9	Minimum Elongation in 4D	Minimum Reduction of Area
Nickel Aluminum Bronze – Full Size	Type I or VI/ cylindrical socket head cap screw	all	96 HRB min	90,000	50,000			
Nickel Aluminum Bronze – Full Size	Type IV or VIII/ countersunk, button head	all	96 HRB min	72,000	40,000			
Nickel Aluminum Bronze – Machined Test Specimens	all	all	96 HRB min	90,000	50,000	54,000	15%	
Manganese Bronze – Full Size	Type I or VI/ cylindrical socket head cap screw	all	60 to 90 HRB	55,000 min 85,000 max	22,000			
Manganese Bronze – Full Size	Type IV or VIII/ countersunk, button head	all	60 to 90 HRB	44,000 min 85,000 max	17,600			
Manganese Bronze – Machined Test Specimens	all	all	60 to 90 HRB	55,000 min 85,000 max	22,000	33,000	20%	

/1 While single shear numbers are provided, parts are tested using double shear. Ultimate double shear minimum limits are determined using the following formula and specific material F_{SU} from the above table:

$D_{RT} = F_{SU} \times 2A$ where D_{RT} = Double shear strength in pounds at room temperature

F_{SU} = Value determined from table, above

A = Cross-sectional area of test piece, in square inches = $\pi D^2/4$

/2 Screws to be electroplated shall have a maximum hardness of 43 HRC.

/3 Unless shown as a range.

/4 0.2 percent offset unless otherwise cited in reference standard. Yield strength testing required only when required by contract.

/5 If proof load testing is required by contract, the requirement for Type I or Type VI alloy steel socket head cap screws is to pass a proof load of 140,000 PSI for parts with nominal diameter up to and including .500 inch, and 135,000 PSI for parts with nominal diameter over .500 inch. Proof load testing is not required for any other material or Type.

/6 Limit per NAS4003, used with permission of Aerospace Industries Association.

/7 Limit per NAS4004, used with permission of Aerospace Industries Association.

/8 Values are for aviation grade Nickel Alloy 718 (Aerospace).

/9 Raw material specimens per 4.3.1.6.1 may be used for shear tests.

/10 Values per AMS4967, used with permission of SAE International (© SAE International. All rights reserved).

/11 Values per AMS5962, used with permission of SAE International (© SAE International. All rights reserved).

5 PACKAGING.

5.1 Packaging. The requirements for packaging shall be in accordance with ASTM D3951, Standard Practice for Commercial Packaging, unless detailed requirements are specified in the contract.

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6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers externally threaded screws, which are designed to be driven into tapped holes with a hexagon key or setter, in applications which require high strength precision products, compactness of design of the assembled parts, used in confined spaces or sinking of heads below surfaces into holes which fit the head.

6.2 Acquisition requirements. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.
b. Type, material, size and length or standard part numbering system (see 1.2, 1.2.1, 1.2.2 and 3.1).
c. Title, document number and date including part number specified in accordance with military document (see 6.3).
d. Material certification, if required (see 3.1 and 4.3.4).
e. Protective coating or surface treatment, if required (see 3.3).
f. Cross drilling of cylindrical heads, if required (see 3.5.2).
g. Thread series and class (see 3.5.4.3).
h. Banded structure inspection, if required (see 3.1.1 and 4.3.6).
i. Corrosion resistance testing, if required (see 4.2.6.1).
j. Dimensional attributes, changes, or alterations (see 3.5.1).
k. Sampling for destructive testing (see 4.2.5).
l. Sampling for chemical analysis (see 4.2.7).

6.3 Military procurement. Items procured under this specification for military use are to be limited to the variety shown on the applicable Military Standards. Personnel of Military departments are requested to refer to these documents for guidance. It is recommended the design of this specification be evaluated for use.

6.3.1 Type I obsolete. Type I (1936 series) cylindrical socket head cap screws are obsolete for new designs. For procurement, it is recommended that the design be evaluated for use with type VI as replacement.

6.3.2 Hazardous materials. Cadmium coatings and coatings containing hexavalent chromium are obsolete and should not be used in re-procurement. Zinc-nickel plating may be used in lieu of cadmium with design authority approval.

6.4 Galvanic corrosion. When corrosion-resistant steel screws are used in contact with aluminum or carbon/alloy steel structures, the corrosion-resistant steel screws can cause accelerated corrosion of the structure due to galvanic effects. In this scenario, it is recommended to coat corrosion-resistant steel screws with zinc-nickel plating per 3.3.5 as this will offer superior corrosion protection of the system than passivation alone.

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6.5 Thread nicks. Nicks on threads are common on fasteners, but can adversely impact torque-tension relationships. Accordingly, manufacturers should strive to minimize the presence of these discontinuities, adding thread protectors when required by contract or at the manufacturer's discretion.

6.6 Subject term (key word) listing.

Button head cap screw
Cylindrical head cap screw
Flat countersunk head cap screw
Socket head cap screw
Zinc-Nickel

6.7 Supersession data. This specification includes the requirements of FF-S-86H dated 7 June 2021. Cross reference data between the types of screws covered by this specification and the types and styles of the preceding specifications, FF-S-86 dated 4 November 1953, FF-S-86A dated 16 May 1961, FF-S-86B dated 23 April 1964, FF-S-86C dated 17 October 1967, FF-S-86D dated 15 June 1971, FF-S-86E dated 29 May 1987, FF-S-86F dated 7 January 2020, FF-S-86G dated 27 January 2021, and FF-S-86H dated 7 June 2021, are as follows:

Table IX. Supersession reference table.

FF-S-86	FF-S-86A	FF-S-86B, FF-S-86C, FF-S-86D, FF-S-86E	FF-S-86F, FF-S-86G, FF-S-86H
Type I, Style 11	Type I, 1936 Series	Type I	None
Type I, Style 12	Type III, 1936 Series	Type III	None /2
Type I, Style 13	Type IV	Type IV	Type IV
Type I, Style 14	None /1	None /1	None /1
None	Type V	Type V	None /2
None	Type VI, 1960 Series	Type VI	Type VI
None	Type VII, 1960 Series	Type VII	None /2
None	None	Type VIII	Type VIII
None	None	Type IX	None /2

/1 These specifications do not cover Type II, Style 11 of FF-S-86 since this type of fastener is an internal wrenching bolt.
/2 This specification does not cover Types I, III, V, VII and IX of FF-S-86E, as these are either 1936 series or have an internal multiple spline socket head configuration (see 6.3.1).

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6.8 Component lot. A variety of production efficiencies accrue from manufacturers processing lots of fasteners part way through the full manufacturing process, and then subdividing the lots further. Some examples would include making a fastener to the longest thread-to-head length possible and heat treating them. Later, to fill orders, parts can be retrieved from that lot, cut to shorter lengths and then plated quickly. In such a case, the lot of parts prior to shortening would be considered a "component lot." Alternatively, parts could be taken to a stage just prior to plating, then held in inventory to satisfy a variety of different final plating requirements. Again, the parts as held prior to plating would be referred to as a "component lot." Documentation for components lot might reflect one lot number, and the finished, shortened/plated lot would have a different lot number. However, the manufacturer's documentation should include a reference tying both lot numbers together.

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL:

Custodians:
Army – AT
Navy – SH
DLA – GS

Preparing Activity:
Army – AT

(Project: 5305-2024-001)

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